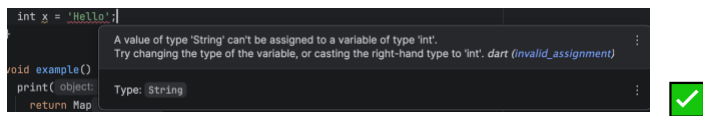
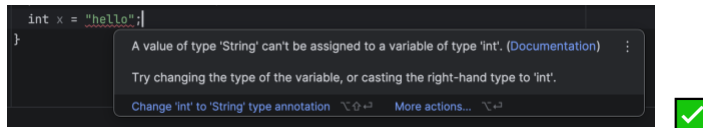
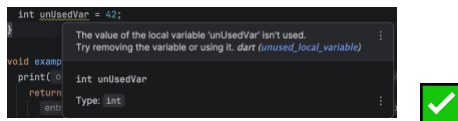
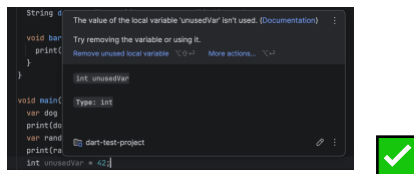


The first image is a screenshot of legacy, and the second image is the screenshot of Lsp with lsp4ij

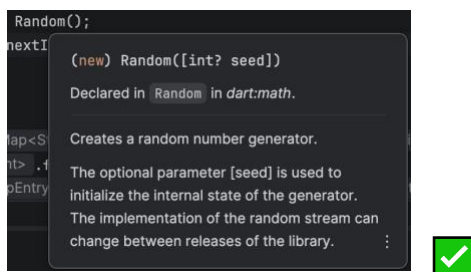
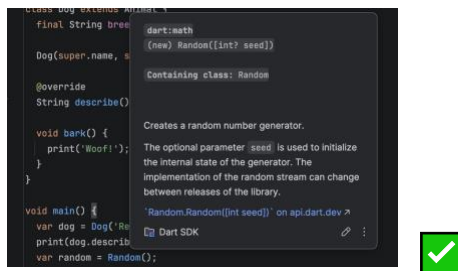
1. Diagnostics - errors



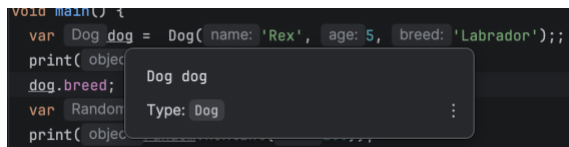
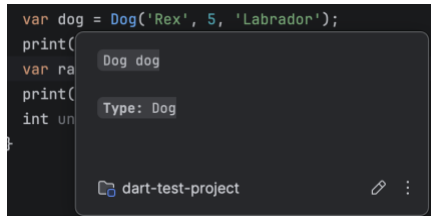
2. Diagnostics – warnings



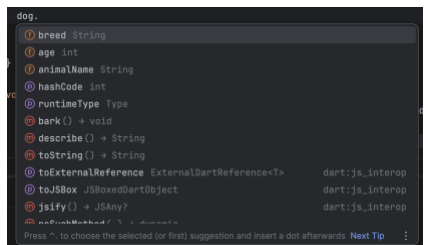
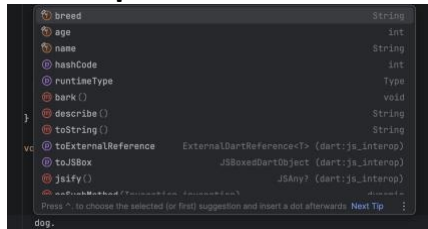
3. Hover - Documentation



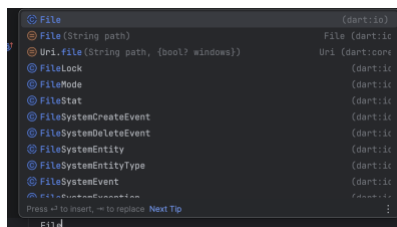
4. Hover – local symbol

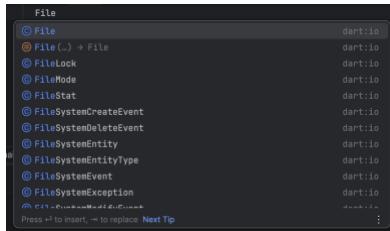


5. Completion – basics



6. Completion – import suggestion





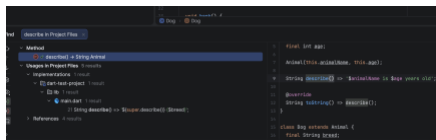
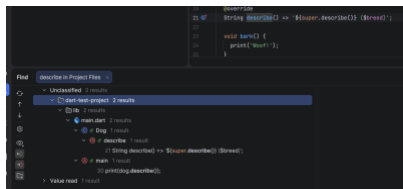
7. Go to Definition – Cmd + Click

- Legacy: Cmd + Click will navigate to definition 
- LSP: Cmd + Click will navigate to definition 

8. Go to Definition – Ctrl + B

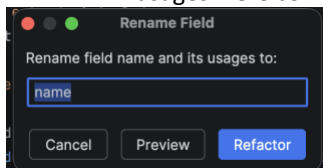
- Legacy: Cmd + B will navigate to class 
- LSP: Cmd + B will navigate to class 

9. Find Usages

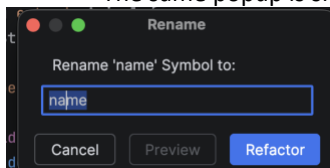


10. Rename

- All usages were correctly renamed to animalName 



- The same popup is showing up and all usages were correctly renamed to animalName 



11. Formatting

-Press Cmd + Option + L

- Legacy: Formatting works correctly ✓

Formatted 4 lines
Show the Reformat dialog: ⌘⇧⌘L

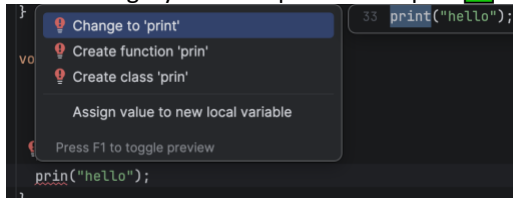
- LSP: Formatting works correctly ✓

Formatted 6 lines
Show the Reformat dialog: ⌘⇧⌘L

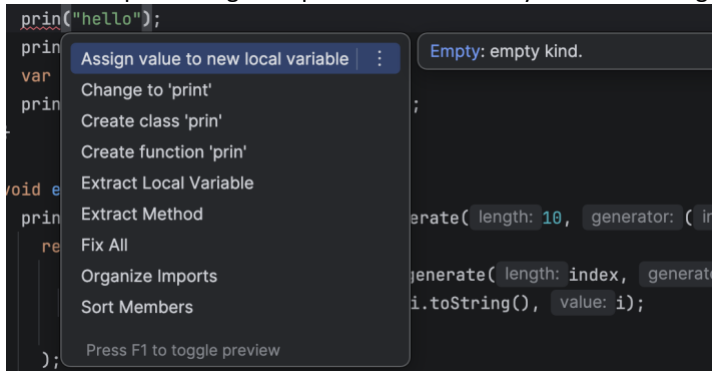
12. Code Action – Quick Fix

-press Option + Enter

- Legacy: Shows options to fix prin ✓

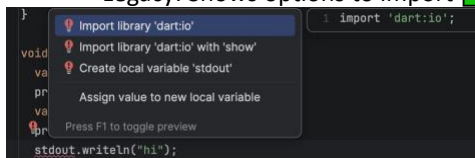


- Lsp: Shows good options but not exactly the same as legacy ✓

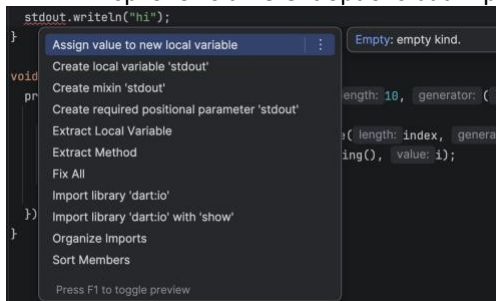


13. Code Actions – Add Import

- Legacy: Shows options to import ✓



- Lsp: Shows different options but import 'dart:io' is not the first option ✓



14. Signature Help – parameter info

```
pr
  name, age, String breed
  Dog()
```



```
pr
  val final String name, final int age, final String breed
  Dog();
```



15. Document Symbols – File Structure

```
main.dart
- Animal
  - age: int
  - Animal(this.name, this.age)
  - toString() = String
  - describe() = String
  - toInt() = String
- Dog
  - bark() = void
  - breed = String
  - describe() = String
  - Dog(super.name, super.age, this.breed)
  - main() = void
```



```
main.dart
- Animal
  - name
  - age
  - Animal
  - describe
  - toInt
- Dog
  - breed
  - Dog
  - describe
  - bark
  - main
  - example
```



16. Folding

- Legacy: Classes, methods, and imports are folded as expected
- Lsp: Classes, methods, and imports are folded as expected

17. Semantic Highlighting

- Both Legacy and Lsp support semantic highlighting but not exactly the same

```

import 'dart:math';

class Animal {
  final String animalName;
  final int age;

  Animal(this.animalName, this.age);

  String describe() => '$animalName is $age years old';

  @override
  String toString() => describe();
}

class Dog extends Animal {
  final String breed;

  Dog(super.name, super.age, this.breed);

  @override
  String describe() => '${super.describe()} ($breed)';

  void bark() {
    print('Woof!');
  }
}

void main() {
  var dog = Dog('Rex', 5, 'Labrador');
  print(dog.describe());
  var random = Random();
  print(random.nextInt(100));
}

```

```

main.dart
import 'dart:math';

class Animal {
  final String name;
  final int age;

  Animal(this.name, this.age);

  String describe() => '$name is $age years old';

  @override
  String toString() => describe();
}

class Dog extends Animal {
  final String breed;

  Dog(super.name, super.age, this.breed);

  @override
  String describe() => '${super.describe()} ($breed)';

  void bark() {
    print(object: 'Woof!');
  }
}

void main() {
  var dog = Dog(name: 'Rex', age: 5, breed: 'Labrador');
  print(object: dog.describe());
  var Random random = Random();
  print(object: random.nextInt(max: 100));
}

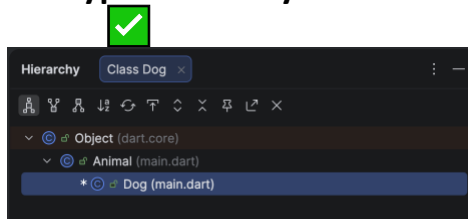
```

18. Implementation – go to implementations

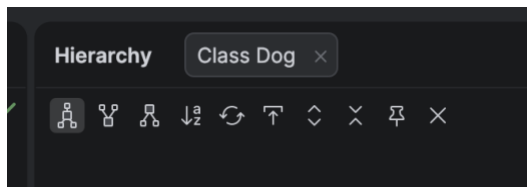
-Place caret on method and press Cmd + Option + B

- Legacy: Goes to implementation as expected ✓
- LSP: Goes to implementation as expected ✓

19. Type Hierarchy



- Lsp: Does not show the hierarchy ✗ (Needs more investigation)



20. Closing Labels (inlay hints)

- Legacy: closing labels come from a Dart-specific custom notification ✓

```
void example() {
  print(List.generate(10, (index) {
    return Map.fromEntries(
      List.generate(index, (i) {
        return MapEntry(i.toString(), i);
      }), List.generate
    );
  });
}
```

- Lsp: Since this is a custom behavior lsp4ij does not have it by default, it needs to be implemented ⚠

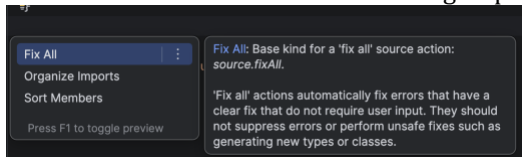
```
void example() {
  print(object: List<Map<String, int>>.generate(length: 10, generator: () int index) {
    return Map<String, int>.fromEntries(
      entries: List<MapEntry<String, int>>.generate(length: index, generator: () int i) {
        return MapEntry<String, int> (key: i.toString(), value: i);
      }
    );
  });
}
```

21. Statement Completion

- Type if (true) without braces
- Press Cmd + Shift + Enter
 - Legacy: Adds braces and moves caret inside ✓
 - Lsp: There is no standard Lsp equivalent for statement completion ✗

22. Sort Members

- Members reordered per Dart convention ✓
- Sort Members is available through Option + Entr by a CodeAction ✓



23. Override/Implementation Gutter Markers

- Legacy ✓
- Lsp: There is no standard Lsp equivalent for Gutter Markers ✗

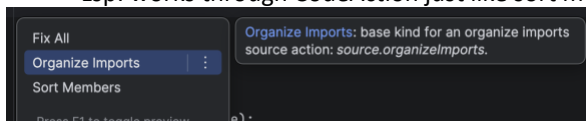
```
@T String describe() => '${super.describe()} ($breed)';
```

24. Go to Super

- Legacy: Cmd + U navigates to super
- Lsp: Dart Lsp server exposes textDocument/super as an experimental capability. So, it could potentially be implemented as a custom handler. ⚠

25. Optimize Imports

- Legacy: Ctrl + Option + O removes unused imports as expected ✓
- Lsp: works through CodeAction just like sort members. Option + Entr ✓



26. Postfix Templates

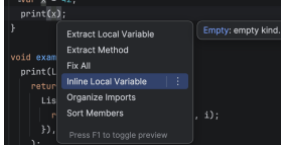
-Type true.if and press Tab

- Legacy: expands to if (true) {} ✓
- Lsp: The Dart LSP server doesn't expose postfix completion capabilities ✗

27. Inline Refactoring

- Press Cmd + Option + N on variable usage

- Legacy: inlines the variable ✓
- Lsp: works through CodeAction. Option + Entr ✓



28. Go to definition – cross file

- Legacy: Cmd + Click navigates to another file ✓
- Lsp: Cmd + Click navigates to another file ✓

29. Go to Definition – SDK source

- Legacy: Cmd + click navigates to source ✓
- Lsp: Cmd + click navigates to source ✓

30. Document Highlight

- Putting the caret on a variable will highlight all usages

```
class Animal {  
  final String animalName;  
  final int age;  
  
  Animal(this.animalName, this.age);  
  
  String describe() => '$animalName is $age years old';  
  
  @override  
  String toString() => describe();  
}
```



```
class Animal {  
  final String name;  
  final int age;  
  
  Animal(this.name, this.age);  
  
  String describe() => '$name is $age years old';  
  
  @override  
  String toString() => describe();  
}  
  
class Dog extends Animal {
```

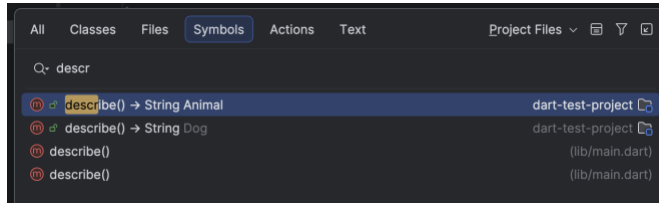
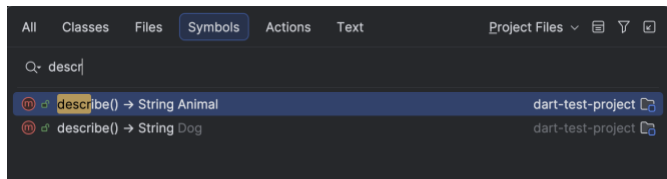


31. Go to Type Definition

- CMD + Shift + B should navigate to type Definition

- Legacy: It does not work ✗
- Lsp: it works ✓

32. Workspace Symbols



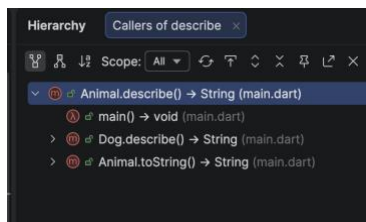
33. Selection Extend

-place caret inside a word and press Option + Up repeatedly

- Legacy: extends selection correctly
- Lsp: works the same

34. Call Hierarchy

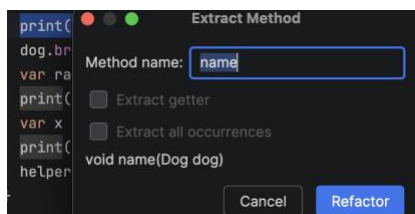
-CMD + Shift + A => type Call Hierarchy



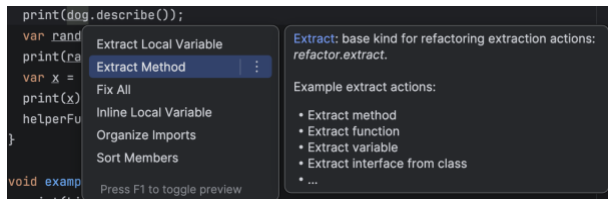
- Lsp: Does not work on LSP (Needs more investigation)

35. Extract Method

-Cmd + Option + M on 2 methods



- Lsp: available through CodeAction. Option + Entr



36. Extract Local variable

-Cmd + Option + V on variable

- Legacy: extracts into a local variable as expected
- Lsp: available through CodeAction. Option + Enter

37. Move File

- Legacy: Moves the files and updates the imports
- Lsp: Does not update the imports. Server supports it, but it seems lspij does not support it out of the box. A handler can be implemented.

40. Cross-file diagnostics

-Rename a function in another file, expect the file that calls it to have error

- Legacy:
- Lsp:

41. Dart Analysis tool window

- Legacy: Errors exist in tool window
- Lsp: Errors appear in the standard Problems tool window

42. Document Link

- Press Cmd + Click on URI will open it in browser

- Legacy: Opens in the browser
- Lsp: Opens in the browser

